

PDF Question–Answer Testing With 'MayurQA'

History - Both Pdfs

1. Q: Who founded the Mughal Empire and in which battle did he establish his power in India?

A: Babur, at the First Battle of Panipat.

2. Q: What was the mature urban phase period of the Indus Valley Civilization?

A: 2600 to 1900 BCE.

3. Q: What was Akbar's policy of 'sulh-i-kul' and what did it represent?

A: An ideal of universal peace and political inclusion across religious communities.

4. Q: Which of the following was NOT one of the major cities of the Indus Valley Civilization?

A: Fatehpur Sikri.

5. Q: What is a common feature in the decline of BOTH the Mughal Empire and the Indus Valley Civilization?

A: Both declined gradually due to multiple interacting factors rather than one single cause.

Geography

1. Q: What is a river's watershed?

A: The land area drained by the river and its tributaries.

2. Q: How can a waterfall form?

A: It can form where hard rock overlies softer rock, where a fault creates a step, or where there is a sudden change in elevation.

3. Q: Name three important Indian rivers.

A: Examples include the Ganga, Brahmaputra, Yamuna, Godavari, Krishna, Kaveri, Narmada, and Tapi.

4. Q: What is a delta?

A: A depositional landform formed when a river slows at its mouth and deposits sediment.

Mathematics - Grade10

1. Q: Solve $x^2 - 7x + 12 = 0$.

A: $x = 3$ or $x = 4$.

2. Q: Solve $2x^2 + 5x - 3 = 0$.

A: $x = 1/2$ or $x = -3$.

3. Q: What is the axis of symmetry of $y = x^2 - 6x + 5$?

A: $x = 3$.

Mathematics - Grade12

1. Q: Differentiate $3x^2 + 4x - 5$.

A: $6x + 4$.

2. Q: Differentiate $x^3 - 6x$.

A: $3x^2 - 6$.

3. Q: Evaluate the integral from 0 to 2 of $3x^2$ dx.

A: 8.

Physics

1. Q: What is the relationship among light speed, frequency, and wavelength?

A: $c = f\lambda$.

2. Q: What is refraction?

A: The bending of light when it enters a material where its speed changes.

3. Q: What does a convex lens do to parallel rays?

A: It brings them toward a focus.

Biology

1. Q: Where does photosynthesis mainly occur in plant cells?

A: In chloroplasts.

2. Q: What gas enters leaves for photosynthesis?

A: Carbon dioxide.

3. Q: What are the two connected stages of photosynthesis?

A: The light-dependent reactions and the Calvin cycle.

Computer — Bits and Bytes

1. Q: How many bits are in one byte?

A: Eight.

2. Q: What does binary 00000101 represent in decimal?

A: 5.

3. Q: What is the difference between bits per second and bytes per second?

A: One byte contains eight bits, so the numerical rates are not the same.

Computer — Operating Systems

1. Q: What is an operating system?

A: System software that manages hardware and provides services to application programs.

2. Q: What is a process?

A: A running instance of a program with execution state and operating-system resources.

3. Q: What is a race condition?

A: A situation in which the result depends on the timing of concurrent operations.